

ActInf Livestream #053.2 ~ "Snakes and Ladders in Paleoanthropology" &

Livestream | & | 2:01:07

to copy or not to copy that is the question from chimpanzees to the foundation of human technological culture recently appeared in physics of life reviews

let me see if this works yeah sorry so it all starts with a gypsum puzzle

and and the puzzle is why a highly innovative species like chimpanzees do not live in a complex technological world so we know chimpanzees are very inventive they are very creative they are insightful they can learn socially but what we don't see in the same extent or similar extent we see in humans is the accumulation of complexity across generations so you can see here in these images we have one of the more complex and demanding tasks that chimpanzees perform cracking nuts open

and they can combine different tools they can use it in the appropriate order so this means they are very insightful and yet if we compare it with a rocket that can bring civilization to mars um it is clear that the accumulation of technology across generation works very differently humans and other non-human species so the the basic question or the one million dollars question is why does innovation fail to translate into technological cumulative complexity in chimpanzees does it owe to a limited capacity to copy with fidelity or what we refer as to imitation which means reproducing not only the end result that would be emulating or emulation but also the steps that lead to that end result so according to this view pillar to copy innovation with faithful accuracy hampers their intra and intergenerational spread and any ensuing ratcheting up effects so plausibly when coping with fidelity is impeded innovative performances are unlikely to live their protagonist and this brings us back to the discussion we had the other day in the snakes and others paper where we said that after two million years ago early homo uh produced this almond chain be facile flake stones and but these um kind of artifacts that we we we don't even dare to call them tools were not copied by conspecific observers because perhaps they were not attributing to them any kind of functional use so perhaps they were not seen as tools and were the result of manual inactions so so to ratchet up complexity it is important that inventions that is tools are copied with sufficient fidelity so that they retain the fundamental features or attributes that make them functional so following the steps of demonstrated tool manufactured by an expert model to the letter is the simplest fastest and cheapest way to make sure that the copies of the original artifact are accurate and here cheapest we refer to in terms of expenditure of energy so if the copies are functional hence useful they keep circulating in the group community longer and this gives increased opportunities for one or more individuals different from the original inventor to add in complexity so for example one individual invents a wooden spur and 100 years later the smelting of copper is discovered and another individual adds to it to this spur a sharp copper point and so on so here is an interesting concept of ratchet effect that would explain in part why humans have built such complex cultural and technological work so the idea of the ratchet effect is that no generation starts

from scratch instead inventions and cultural products created by one generation are kept in mind and transmitted to the next generation this way every generation is better off supposedly than the one that preceded it so here we can think of books books are a wonderful way of keeping information sharing it across generations and among the same generation also so thus technological complexity and success in humans

is not dependent solely on the geniality of one or few individuals it is a joint endeavor where many individuals contribute their specific expertise and skills to produce increasingly complex artifacts and again we can think in um establishing a colony in mars no single human alive could achieve that alone or would have

the required expertise to achieve that alone but all together we can perhaps manage that so we need this kind of shared knowledge and accumulated across generations of course

so why it is some behaviors or innovation are copied or not can be explained at the level of social dynamics but also

at the level of the brain or giving a more mechanistic explanation so explanation why copying of shared behavior in chimpanzees does not occur with frequency have been often kept at the level of social dynamics

where factors such as social prestige dominance etc are said to be key to explaining whether or not an invasion is adopted by observers

and this approach relegates the brain to a subsidiary role

here michael and i will be attempting to identify what the minimum cognitive requirements are that can give rise to innovations

and their horizontal and vertical transmission

and we also attempted to differentiate the cognitive ingredients necessary for innovation from those that need to be in place for the adoption of an innovation

so we wanted to establish a clear demarcation between the brains of the

ones that innovate and the brains of those who observe an innovation because we think this is important to be analyzed separately

so in order to see which are the mechanisms at the level of the brain that would allow for one innovation to be copied we invoked the lens of the free energy principle according to active inference under the free energy principle it is through there being limited to a restricted number of states that is the attracting set of the free energy principle that organisms minimize the variation of free energy of the internal states and resist entropic dissipation minimizing variation of free energy reduces prediction sensation discrepancies prediction errors that can be corrected by active sampling of predicted sensory inputs and perceptive modification of predictions this is sometimes expressed in terms of perceptual inference that is changing predictions to match sensations and active inference reactively changing sensations to match predictions so

innovation in a way might represent a potential misfiring from other winning perspective

so minimizing variational free energy can be achieved reaction sent in creatures can act

active inference in order to fulfill their predictions and thereby reduce discrepancy between the latter and

sample reality

being limited to a restricted number of states what we could refer to as the attracting set is therefore

synonymous with minimizing variational free energy and hence maximizing adaptive fitness

here innovation represent an anomaly and innovators are unorthodox actors or protagonists who entertain new courses of action or plans

new phenotypical responses that have not been previously entertained

and according to this interpretation of the free energy principle it is at the level of policy selection where innovators stand apart from the crowd their policy selection is in a sense faulty insofar as their ensuing behavioral outcomes are in mark contrast to those of their fellow actors who deploy orthodox responses that reduce discrepancies between predictions and sensations thereby reducing the prediction error

and here we will talk of miss firing we don't do it in a kind of negative sense it's just to express that these new behaviors these innovations necessarily will produce more surprise

okay seeing surprising things with this phrase i i'm kind of paraphrasing and the clark in a paper in behavioral and brain sciences he referred to seeing surprising things to being able to lend credibility to the sensory information that is not in accordance to prior beliefs so the family of probability

probability density is entertained by an individual is restricted to the systems the brains prior bias and beliefs equivalent roughly speaking to its phenotypical repertoire

this means by definition that innovative behavior is both unlikely to be evinced in any given phenotype and crucially difficult to recognize when observed by another

should an innovator's behavior diverge markedly from the repertoire plausible policies used usually by an observer

to predict another's behavior it may be beyond the capacity of the latter's generative model to recognize it that is to make sense of it so large discrepancies pose a threat to the brain because one updating involves making energetic demands and two risk reducing adaptive fitness by inappropriately changing changing the generative models and here we need to think that the generative model has been selected also in a specific econo so changing it might might have risks okay

so what is required for a brain to entertain generative models that involve unusual or bizarre behaviors the hierarchical predictive interpretation of the brain or hierarchically mechanistic mind

an inference inference machine sorry implies that each level of the hierarchy of cerebral neuronal circuits conveys predictions or conditional expectations that suppress errors at the level below

while concurrently conveying errors that permit the revision of the expectations at the level above plausibly dynamical precision weighting enables the system that is the brain

to readjust the confidence afforded to top down expectations as opposed to bottom-up prediction errors in other words in order to entertain a prior that an unlikely hypothesis might explain best some behavior witnessed it is first necessary to relax the precision or commitment to prior beliefs

thus enabling new or implausible hypothesis to explain novel behavior and here i introduce

some things we discussed the other day professor friston talk of attenuation of sensory information in relation to saccades which is kind of switching off this sensory information we are being aware of and that

would be the opposite as being impressionable so

i'm quoting professor friston here if i can copy that means i am impressionable i will accumulate new evidence very quickly

i have the capacity to assign greater evidence or precision to sensory evidence and suspend the precision to my prior beliefs

this is something he mentioned in the last uh active infant recording 53.1.1

and to end this introduction one concept we introduce in our paper which is central to our discussion is that of zones of banded surprises so this is a new concept introduced to refer to the levels of mismatch between prior beliefs

or top-down predictions and sensory information that a species can tolerate

so first and foremost the focus of the zbs concept is on the observers and their ability to make sense of unobserved innovation

and coding and subsequent copying of innovation depends on the degree of disparity or surprise between predictable behaviors namely those typical of the species and the perceptual incoming information about novel behaviors if the zbs is wide it means that the species can give credit to the information received through its senses that challenges the prior beliefs in short a species can register encode and potentially reproduce reproduce in the behavior of certain code specifics that deviate markedly from customary prototypical behaviors the species is highly impressionable if we use the words that professor

friston used the other day so if the zbs is narrow then a species ability to change prior beliefs to accommodate new evidence regarding innovative practices observing goal specifics is compromised

i just to end up and avoid any kind of misunderstanding want to make clear that this our sound rebounded surprise have nothing to do with the zone of latin solutions by tenure i'll be we are discussing similar topics so the son of latin solutions that was introduced by tenure kai and tomo waselo refers to

solutions to a problem for instance extractive foraging that can be achieved by an individual in isolation during his lifetime and hence is not likely to be the product of cumulative culture involving social

learning when observed in the wild and for those readers interested in this concept they can go to the

original paper i'm i'm showing here and also there is a recent paper by andrew whitehead who is professor in in saint andrews that discusses this concept and that has replication by other authors so it might be a good update for those interested also and with that i finish this short presentation and we can enter

discussion let me stop sharing screen i think okay now thank you hector i will bring up the notes again and to michael carl or dean whomever would like to provide a first response please

well i don't really have a great deal more to add um i think

hello yes please continue what's happened here you're all good michael we're here michael hello can

you hear me yes we hear you continue yes we can i'm not quite sure what's happened here we seem to have snakes and ladders up on the up on the screen um as i said i i don't think i can add much more to what

hector has just said i think it makes a very succinct summary of a bar paper and

i would like to certainly congratulate hector for making what seems to me a very original and perceptive observation that it is the observers whom we have to thank for

uh copying and therefore for the handing down of technological achievements by early humans whereas too often the emphasis has been made on the person responsible for making a technological innovation uh but of course unless somebody copies that or at least unless it registers in the mind of somebody else it is unlikely ever to get copied and although this seems to be a fairly a fairly obvious observation it seems to have escaped most of those who have published on the matter hitherto and i think this is a very original contribution on hector's pot which deserves uh to be complimented and uh uh really that's all i wish to say i think i'd like to throw this open to other people and if i can illuminate or elucidate any matters i'm happy to do so but i think this is really very much uh to put it mildly hector's brainchild and uh congratulations

thanks michael you're very kind but this is our child so and you you know a lot you have a lot to contribute so i mean i can only learn so but thank you anyway awesome carl or dean

go ahead carl right so i just wanted to um reiterate michael's admiration for that for both the idea but also the very fluent um and clear articulation of it i was um it was nice for me to see that uh presented so succinctly because the clarity of the ideas really um is really very evident um it struck me there is sort of two um two themes that we could discuss which is the um the emergence of more complicated solutions to accurately engaging with and modeling our world i think that was a generic theme in the first part of the presentation but then very quickly we moved to this kernel idea this core idea um that it the dissemination and the transgenerational propagation that um could be read as some kind of um cultural niche construction in a sort of evo-devo context depends upon being able to recognize a particular behavior uh and thereby copy and imitate i think that that's that's that was the key thing and for me um just understanding the mechanics of that from the point of view of um belief updating and active inference um was a i think a really useful thing and as um hector beautifully uh articulated uh articulated the ability to recognize something new and update one's generative model or posterior beliefs depends very sensitively on the confidence or the precision that you ascribe the sense of evidence in relation to the precisions with which you hold your your prior belief and i think that that's absolutely crucial and i i sort of um highlight that issue because it is exactly the same issue that you will find when applying active inference and the free energy principle to um psychiatry and the different ways in which we can uh i can't remember hector what was your word misfire uh in neither a good nor a bad sense but certainly there are certain psychiatric uh syndromes that could be regarded as misfiring and of course there's a very close association with innovation and creativity and manic depressive psychosis as you know uh you know famed in terms of um various historical figures in uh in the arts um and that i think is you know again brings us back to this key notion of getting that balance the balance of the precision of our prior constructs what we have accrued through experience dependent learning

and the precision afforded evidence that this is how the world is actually working and how i am actually working getting that balance right

um and that's a really difficult job and sometimes in computational psychiatry we um we try to explain this in terms of a statistician

um in order to get the balance right you have to be able to estimate these precisions

and this exactly the same spirit that a statistician when making an inference doing a t-test has to estimate not just the difference in the group means but also the standard error and it's often much more difficult to estimate the standard error

than it is to estimate the difference in in in the group means and that standard error is the inverse precision it's the weight that you afford the meaning the significance of some quantitative difference say a prediction error

um so you see this theme that you know there are two parts to inference the first order part is just the prediction error or the surprise but then you have to assign how newsworthy is that you have to actually modulate that with its precision so you can multiply by the precision or divide by the standard error to get the actual

thing that matters that's going to change your mind um and there's a whole literature on um different kinds of people in the context of neurodiversity that i have an impairment in being able or possibly have an in being able to estimate that precision and i'm thinking here of the the poster child of this kind of formulation which is autism um and the idea being that there is an early developmental failure to attenuate sensory precision so they are rarely and continually during their early development severe people with severe autism um unable to ignore any of their sensations um and as a consequence everything is surprising everything is arousing everything is in danger of revising any um beliefs very very quickly so they're highly impressionable to the extent they can't build deeply coherent models of their lived world or indeed the other artifacts such as mum uh that constitutes their world so i i'm just wondering whether you know a species that um has the same kind of latitude in terms of either biasing attention or precision the sensory evidence versus accumulated prior beliefs would display the same kind of misfiring so you can go from one end and autistic species will be overly impressionable it'll be too evolvable it won't be able to remember anything from generation to generation the converse is a um somebody who you you you can imagine from a psychiatric perspective um has committed unduly to very precise

prior beliefs about the way the world works and you can think of this in terms of compulsive obsessional disorder or perhaps even depression you know the world is an awful place that is true therefore there's no point in going out and you know challenging the evidence that the world is an awful place so i'm just going to stay inside and feel um um isolated and not move or say very much um so that would be an example who somebody is not very impressionable in fact their pathology their misfiring prevents them from acquiring any evidence to revise their beliefs

so if you imagine a a species um that had that same kind of um second order misfiring on this so this is not the misfiring of the innovative so this is a a failure to infer the second or cystics the reliability the precision um the variance the entropy of the content of the things they're trying to

innovate or or make sense of you you're you're left with this notion of a a species that just cannot respond to selective pressures for selectability they can't respond um by adjusting their evolvability simply because they're fixed in their in their way so for the environmental eco niche changes in any fundamental way they're not going to be able to um to respond to that um so i think this you know there are many examples of this sort of key um uh this key if you like yes i think it would be called a pathology in the sense that in a volatile environment um you would need to optimize the precision afforded sensory evidence in order to make the best sense and understand and engage with that environment and if you get that wrong there will be a pathology um and i think basically hector's um uh formulation really drills down and really focuses on the actual mechanism this kind of um this kind of uh pathology you know it's all about the transcultural spread of generative models and and sense making um in terms of engaging either with artifacts or indeed with you know with conspecifics and i repeat it's such a such an easy thing to get wrong as as evinced in in psychiatry you know there are so many instances of um neurodiversity expressed clinically that i think reflects this this singular dimension in terms of the ability to get that balance to get that balance uh right i remember we discussed last time again in terms of impressionability and uh selection for selectability and the like the same kind of mechanism unfolding uh you know in terms of evolutionary uh mechanics but at the level that i think we're talking about today in terms of um you know evincing plan uh pursuing plans um and novel ways of doing things that may or may not be um recognized uh due to the due to a failure or not of getting um being able to suspend the precision of your prior beliefs i think that that was a central um central theme i'll just close um with an interesting comment that in terms of the kind of misfiring that um hector was implying basically a novel way of doing things um being the unique province of those kinds of creatures that can plan possibly deeper into the future um there's a really interesting sort of link back to this precision notion in the sense that if you try to simulate the depth of planning of any sentient artifact then the degree to which they are able to look into the future depends upon the precision of their beliefs about the succession of contingencies what will happen if if if this happens and then what will happen if that happens so just technically the precision of the transitions amongst worldly states of affairs as you roll out into the future and again you have to estimate that precision um so there's another if you like point at which precision gets into the game but in this instance it's not the precision of the sensory evidence but the prior precision about the ways things unfold and what will happen if i if i do that if you have a low precision there then you don't plan very far into the future because you lose certainty and confidence very very quickly after just a few moves whereas if you have very precise beliefs about the um the state transitions and the consequences of various actions or sequential policies then you can roll out confidently much further into the future and perhaps find that innovative innovative solution so again we come back mechanistically to phenotypes that have differing degrees of precision that um the in this instance get into the game in terms of the um bounding the um bounding the the the depth of the planning or the temporal horizons um that you can entertain um when um when um scoring different plans in order to actually select the best action at the moment

so for me that insight is really useful because i can see how one might want you know how one could simulate

that quite easy easily in sort of toy situations so that those are my thoughts inspired by by excellent hector's excellent uh presentation

thank you carl dean what say you

well if it's if it's okay i'd like to sort of step back for a second and um

um because we were privileged to be able to look at the two papers almost simultaneously i know chronologically we've we've split it up into weeks but as readers we were able to kind of look at

both paper side by side i kind of want to take a bit of a hofstetter strange loop two-sided condition mobius we've got that all those representations in our in a lot of our live stream introductions

so if we were to assume that because carl pointed out not everybody can accommodate a two things at once perspective if we can assume that that doesn't overwhelm my question to the authors is do

they see their own journey as a one paper or artifact artifact morphing into the next or and as we were pulled out and taken up by this relationship between epistemic and pragmatic values when up

taking an active inference perspective that's part one of the question part two is the former is kind of a sequential mechanical technical side of things the latter is the combinatorial plus meaning entailed

and entangled so more more where am i relative so combined that's a relative depth of field as

setting as establishing morphing we want to call it that as active inference strategy is applied how do

the authors see or potentially see this perspective combining meaning the sequential and or the combinatorial affective effecting not effecting but affecting what we believe because i'm a big i'm a big believer that

we can't just sort of strip the feelings part of this out i think that's whether we're talking chimpanzees or whether we're talking the emotions of sliding down a snake those are those are part of this thing that

we're looking at i'll just give one short note on that in the paper sophisticated affective inference

they simulated counterfactual futures projections about futures which are inactive inference action oriented all futures are analyzed and inferred about with respect to policy selection by the agent

and one of the patterns already apparent in even the simple simulations were that even if a very small fraction of counterfactuals have like death as an outcome or highly negative outcomes it can increase the variance

estimator on possible futures greatly leading to essentially ruminative behavior and affective anxiety

uncertainty around the legitimately warranted variance in possible outcomes so i see a key piece of this puzzle

to be entertaining progressively wider time cones of possible futures in a prospective capacity

and entertaining progressively wider time cones in the past of causal mechanisms

and somehow negotiating that affect barometer in the moment as the eye of the needle

hector michael where do you see affect coming into the snakes and ladders to copy and not to copy set the

uh well may i make one or two comments um first of all in addition to what professor friston has told us

about autism and so on i think it's worth mentioning uh in the connection of the question that daniel has just

raised to say that there is clearly a difference in the evolution during the life cycle of the early humans on the one hand and great apes and probably most human forerunners on the other the extension of the human life cycle the extension uh beyond childhood and in particular as i mentioned last time

um the adolescent growth spurt which we have but which may not have existed uh in early homo uh even one and a half million years ago i'm sure is relevant to making the evolutionary uh step between let us say the surprise that perhaps a chimpanzee has if she observes a dead bird and then regards this as counterintuitive

uh because birds ought not to be lying on the ground throws it up into the air but this is a an individual spontaneous uh reaction which is not um handed on as part of a behavioral repertoire amongst the species and on the other hand when we begin to see the reproduction of as it were counterintuitive uh behavioral sequences such as uh taking a lot of flakes repetitively by hand from a piece of stone using another stone to produce an almond shaped hand axe fairly regularly flaked on both sides something which is not found before homo appears in the uh archaeological and paleoanthological record but which is not something that seems to be a regular part of human behavior it is almost as if uh the individual who made that uh was doing it as a kind of uh spontaneous thought of what might happen uh if i was to do this uh part of the deep generation uh of the uh hierarchically mechanistic mind which professor fiston has told us about a deep generation that however may have been a misfiring it may have been simply a personal spontaneous event which is not easily uh recognized by others particularly if they're uh shall we say to put it mildly their mental uh development is stunted due to the early adulthood and the lack of any adolescence adolescence in particular uh which and uh of course um the development of the neuronal architecture of the brain which in modern people continues until our mid-twenties if this was not to have occurred uh um as it does not of course in um even in great apes then there would have been perhaps a problem here about the deep generative potential of an individual ever being uh recognized such that um it isn't just what might happen if i was to do this with a piece of stone but what might happen if i was to copy what that other individual is doing with a piece of stone um and this i think is the stumbling block this is what has uh led to the snakes and ladders picture which is what i think has led to a very slow adaptation of technological uh accumulative technological um uh handing down from between generations or between individuals of a generation and between generations and it may well have been the case that it was not until dem uh demographical as well of course as uh biological paleobiological changes had evolved in the genus homo that there was sufficient uh intercourse uh not merely sexual but social uh such that um these questions continue to be uh asked ever more frequently and with the appearance of better and ever more fluent communication uh would allow uh what we now uh regard as human culture um specifically uh human nature uh which uh i think uh differentiated has differentiated us ever more from uh the the great apes um i just those are my thoughts on the matter um

thank you michael carl or anyone else please

can you please um

can you please um speak to the question about affect and what michael has just treated as two um

so one way of um conceiving of affect and valence and um you know the emotional aspects of choosing different plans or um making sense of my own bodily sensations um is to think of it in terms of precision estimation um so one way of carving um one way of a very generic way of thinking about some negatively effectively charged states of mind would be those states of mind in which there is irreducible or unresolvable uncertainty so you know to put simply um to be uncertain about what is going to happen to me or what i should do is to be in a state of angst um and that can be um linked to notions such as anxiety um and you know with some extra nuances other um feelings such as you know um a low affect and indeed if that is pervasive a low mood or or depression um but i think the key thing here and indeed many

people uh have developed a calculus of emotion based upon whether your average level of uncertainty is increasing or decreasing or is high and low and adding these things together to get a an ontology of emotional states from a purely sort of inference perspective there's another a body of thought and um literature which links emotions to um the literally gut feelings interception um so the

certain emotional states are just hypotheses about the way that i am feeling um now the best explain the myriad of extra receptive interoceptive sensations so i think there is a mechanics of affect in um

um certainly interceptive inference um and possibly more generally in terms of active inference

however there is another aspect of um things that we the the feelings that we might associate when we feel um compelled to do things and i'm thinking here not about the irreducible um or irresolvable

uncertainty but about uncertainty that can be resolved i'm thinking here um when hector mentioned

um andy clark's treatment of surprise there is also the truism that we expect to be surprised um and

technically that lends um that can be explained in terms of the imperatives for choosing those actions that will resolve uncertainty um about the way the world works so this is just an expression of novelty

so if we anticipate if we if we anticipate that we are going to be surprised um by some outcome what we are saying is that that outcome is going to be informative and that it will reduce some uncertainty

technically it has a high expected information gain so i'm speaking here as a statistician that this

and using terms that would underlie uh the bayesian principles optimal experimental design that we

we do these experiments uh specifically in order to get surprising outcomes that are by definition

going to resolve my uncertainty about my different hypotheses so if we think about the

the uh the misfires that are innovations these are effectively by definition novel behaviors and they

are um the fact that people um or certain people and certain creatures engage in um novel um in novelty

seeking is entirely sensible from the point of view of the free energy principle because novelty um seeking just is an expression of curiosity it is just an expression of choosing those outcomes that are going to yield

rich information that has um technically a high self-information or an expected information gain so

that this is another view of the um if you like the effective aspect that we do we are compelled and expect ourselves and will possibly explain ourselves as being excited um in a positive way and the opportunity to resolve

uncertainty because we see an opportunity to do something novel what is behind that door you know what is inside this present

you know what would happen if i did that and that's an entirely natural um and by natural i mean something that is easily naturalized under the physics of self-organization of any system that has or entails an implicit generative model of the consequences of it of its own action so i thought there was a lovely link between that kind of um um affect written into the um the the imperative to seek novelty to seek surprises in the future um and what michael was saying about uh well yes in all the order to do that you still have you still have you have to suspend your um your prior convictions because if you a priori think you know everything there is no novelty there is no imperative for expiration uh and you can just do the same thing time and time again but if you're in a if you can recognize that there is reducible uncertainty and um choose those actions that have the greatest expected expected um expected information gain then you will necessarily succumb to um novelty seeking behavior and thereby discover those misfire innovations so uh and it may be that the you know there is uh well the story that michael was telling uh was that we are built to be curious creatures for much longer we are built to you know to be able to respond to novelty and to find those innovative novel ways of behaving um uh um you know so that we remain um we we retain that childlike curiosity um as part of our epigenetic epigenetic neurodevelopmental programming for much longer than other species so i i think that sort of um matches very nicely with the sort of the basic thesis that hector was proposing during the presentation it all comes back to getting the precision of your prior beliefs right and suspending um very precise prior beliefs in order to explore hypotheses and that exploration just is in terms of engaging in novel innovative behaviors that will um that you know will resolve your uncertainty so i i if that's right it sort of engenders a question why is there an evolutionary pressure for this kind of eternal childhood so um i want to ask michael and hector because michael certainly brought in um language and cultural niche construction i i i my guess would be that there will be a circular causality um between um the the the cultural eco niche and the phenotypes are trying to adapt to that cultural eco niche such that they are required or the the phenotypes that have the greatest adaptive fitness are those that retain a greater curiosity um in the context of the culture in which they find themselves and part of that will be necessary to actually understand the communication and the signs and the semiotics afforded by their conspecifics which could be read just as language so my question is did this innovative does innovative tool use does it emerge um um stochastically or um you know um at certain points along with communication and language um or um are they

or are they not correlated and if they are correlated in terms of you know um you know historically um what are they are they're thinking about the direction of causality does the fact that you um remain curious and innovative for much longer um lead to language that we can you know engage in novel communicative behaviors or is it that we need to um remain curious and open-minded in terms of um thinking about or entertaining novel plans and innovative plans simply because we have to make sense of a world that is rich in information because there is language that is that has been culturally inherited or is it both is

there is a both top-down and bottom-up causation so so two questions language tool use and copying and the um the coupling between the sort of evolutionary and uh phylogenetic um sorry the phylogenetic um the um the um sort of phenotypic uh levels when it comes to things like language well um i wonder if i could come in here uh daniel is it okay if i address uh cause questions yes in whatever order you like

um thank you um well in the first place there is a voluminous literature on the relationship between language and the origins of tool making and tool use uh really and there's a lot of uh unresolved argument about whether the two came together very early in human evolution let's say between one million and one and a half million years ago, as I originally thought some years ago, or whether in fact language as we know it, or shall we say fluent communication at least, only developed much later, as I'm coming round to believe now, and as Noam Chomsky has said for many, many years.

I don't want to get involved in that because really we would need two or three programmes and a lot of literature to compare and contrast.

Rather, I would query whether we have an eternal childhood.

On the contrary, I think that physiological childhood is slowed down and when the adolescent growth spurt arises, after about the age of, let's say, nine years old, at some times or perhaps somewhat later at others, is when we see a major change in behaviour which is no longer childish, but neither is it adult.

Of course, we needn't go into that.

We all know that adolescents can be very impetuous, but they can also be extremely anxious.

And this brings us to the angst that Karl has mentioned, and which I certainly felt as an adolescent facing the weekly Viva Voce examinations in human anatomy in the dissecting room, which in those days we all had to undergo, at least in the United Kingdom, as I can sure call the members, and which always put me into a state of extreme nervousness and has given me a lifelong worry about the lack of accuracy, let alone precision, in anatomical description and identification and a concern with the statistical variability in human anatomy or aspects of it, or anatomy in general, not necessarily human, and its relation to process,

in other words, to physiology.

And I feel myself much more,
despite having lectured in human anatomy,
much more aligned to physiology.

And I spent some time doing neurophysiological stuff
many, many years ago,
at Oxford and in Göttingen,
than in the classification,
this has to be that way.

And as we were taught, as medical students,
always think doctor about the most obvious
rather than the most obscure.

But we have to have the whole range before us.

And this brings in the whole problem of,
do we think in terms of very right or wrong,
concrete facts about stuff in general?

Or do we think in terms of the statistical range
of variability,

whether it is in terms of occurrences,
occurrences,

let's say related to Poisson-type distributions,
or to continuous variates,
which can be, let's say,
related to Gaussian-type distributions.

And unfortunately,
it was only much later in life
that I discovered that, in fact,
modern statistics allows these two approaches
to be compared and contrasted
in a uniform manner.

I didn't know that 50 years ago,
alas,
which shows my ignorance.

Still,
I would come back to the whole business
of the growth curve of humans.

This clearly has altered.

When we look at chimpanzees

and other great apes,
basically,
they don't have much of anything
approaching our adolescence.
They don't have this slowing down of growth
and then a recuperation of the velocity of growth
combined with,
most important,
not merely anatomical and physiological changes,
but also those of the mind.
The ability to innovate,
the ability to have lateral thinking,
or the ability,
which I have noticed in some young people
I've met with Asperger's syndrome,
to be able to concentrate
in great detail
on particular matters
which are extraordinarily arcane,
such as Sanskrit literature
in people
that are not of a South Asian heritage,
whereas
in dealing with other people,
their other colleagues,
they appear to be,
shall we say,
gauche.
And this is, of course, well known.
However,
and indicates the range of variables,
but it is the change
which we all undergo
in not merely physical,
but also intellectual ability
and inquisitiveness,
which can get us into trouble
when we are in adolescence

and as young men and women
given the
continual
and continuous changes
in our neuronal architecture,
at least until our mid-twenties these days,
which I think distinguishes
and has distinguished
the evolution
of Homo sapiens,
well Homo in general
and Homo sapiens in particular,
but I wouldn't rule out Neanderthals
or even Homo erectus,
and which I think
has been cumulative
and undoubtedly,
although we have no idea,
undoubtedly
enhanced
by
increased,
excuse me,
and ever more complicated
communication
during the past
probably half a million
and certainly
a quarter of a million years.
This curiosity,
what would happen
if we did this,
what would happen
if we did that,
but
even so,
things are not
rapidly

grasped.
and
we see
that in different parts
of the world,
particularly in modern populations
even that have been
long separated
from each other,
not everything
is developed
at the same rate.

A couple of well-known examples
are, for example,
the wheel,
which first appears
probably
on the Eurasian steppes
about 5,000 years ago
and was used for traction,
and then appears
quite independently
in pre-Columbian
Meso-America
about 2,000 years ago
and was not used,
even though it would have been
very useful,
for even anything
as simple as a wheelbarrow,
but simply
to move around
images,
whether they were dolls
or toys
or gods
with no idea.
and another one

taken from the same
two areas
of the world
is the metallurgy
of copper,
which appears
about,
well,
perhaps over 6,000 years ago
in Eurasia
and possibly
2,000 years ago
independently
in the New World.
and yet
other things
happen too.
For example,
in Australia
we find
modern people
were making
hand axes
when these
had long ceased
to be made
by anybody
outside
Australia.
Some of them
may even be
only a few
thousand years old.
so
this
appearance
and
disappearance

and reappearance
is something
which
has not
gone away
and must
be related
somehow
to
our ability
to
this
deep
generation
of
our inquisitiveness
our curiosity
and indeed
our ability
to
not to see
the most
obvious thing
around us.
The desire
to see
only
one
simple solution
which so
irritated me
when
my teachers
in the
anatomy department
when I was a
medical student
believed that

there was only
one
right answer
as to
how many heads
this muscle
had
or
how many
vertebrae
there were
in this
or that
part
of the
vertebral column
let alone
in the skull
and we know
how very
variable
these matters
can be.
so
there is this
tension
between
the desire
for things
to be seen
in one
light only
in a
one dimensional
way
and
our
natural

inquisitiveness

this

deeply

generated

desire

to say

what would

happen

if it was

like this

how has it

come to be

like that

why is

everything

different

and not

the same

I can't

put it

any other

way

with that

I've said

quite enough

thank you

Michael

Dean

thanks Daniel

so again

this is for

the authors

this is

this has been

really helpful

to me

as observers

because that's

that's
highlighted
very
clearly
in the
papers
I see
both papers
speaking to
something that I
would describe as
condition set
meaning
in the papers
examples
limit the game
to
the game
metaphor
at least
to
this type
where there's
a limited
number of tiles
and a
clear
continuation
until a
conclusion
or
recognize
the limit
of the
species
to
nut
cracking

so
there's
a real
clear
idea
and
Michael
and Hector
have both
spoken to
so
what was
the
growth
limit
of the
brain's
capacity
etc
what I'm
curious
about
and I
don't know
if there's
Michael
or Hector
have a
real
clear answer
to this
but
I'm
wondering
what
and when
was
the

was maybe
the range
of time
that we
see
the
unlocking
or
or
or
sort of
combinatorial
explosion
or not
what I
guess what
I'm asking
is
is there
a move
that we
can
clearly
see as
being
something that
differentiates
playing which
we know
happens in a
lot of
species
and gaming
because as I
said at the
end of
53.1
I wondered

if it was
going to be
game on
game on
and so
I'm
curious
what the
authors
think about
is there
a difference
between
playing
and gaming
meaning
that
there's
now
an actual
introduction
of
strategy
in this
so
maybe
another way
of asking
it
is there
a moment
when you
guys can
point to
it
and say
this was
kind of

when we
saw
a reducing
or maybe
a relaxing
of a
limit
forward
world
I wonder
whether in
fact
this difference
is really
quite as
great as
Dean suggests
and whether
it may be
have something
to do
with the
ability
in
working
memory
and
other
aspects
of
memory
in
dealing
with
or in
developing
logical
mathematical

ability
to
plan
or in
planning
and in
communicating
this ability
such that
the planning
is no
longer
on an
individual
level
but is
participatory
and therefore
again
in other
words
and perhaps
Hector can
answer this
question
but I would
suggest
that
this
developed
slowly
gradually
and
by no
means
in a
continual
let alone

a continuous
manner
in Homo
but not
in earlier
in a way
not in
earlier
and other
hominid
species
do we see
and this
I would ask
Hector
do we
really see
let us say
chimpanzees
or other
great apes
developing
the
kind of
logical
math ability
to make
very very
simple
combinatorial
associations
between
objects
or
behavioral
activities
and
developing

this
into
a communal
game
which involves
other individuals
in the group
because
as Hector
has remarked
on various
occasions
and informed
me
when we
see things
like
nut
in
small
groups
of
chimpanzees
it by
no means
is copied
by all
of them
let alone
copied
or emulated
by all
of them
and by
no means
even when
it is
with

faithful
accuracy
and precision
in the
manual
operations
that are
required
so
that
I would
ask
Hector
do we
see
these
associations
being
transformed
into
games
okay
that's a
tricky
question
I mean
what we
know
with
chimpanzees
is that
they have
promised
to do
these
second
order
relationships

which is
like
hierarchical
ordering
of
for instance
if I have
a set
of objects
I can
classify
them
attending
to the
color
for instance
chimps
could do
that
okay
if only
color
is involved
but I can
classify
them by
color
and shape
so this
is another
order
of
hierarchy
so I can
pick things
that are
square
and red

and put
them
together
so this
ability
to
abstract
a kind
of
ordering
category
so that
I have
relations
of
relations
like in
analogical
reasoning
when I
can focus
on one
dimension
or in
another
simultaneously
so I can
shift my
focus from
one dimension
to another
in order to
classify
things
this I
think
is
quite

human
like for
instance
the Wisconsin
card
sorting
test
so I
can
shift
from one
dimension
to another
to classify
objects
or cards
and this
is something
that teams
would have
more problems
to do
so they
might
classify
in a
first order
categories
like I
put things
that have
a same
color
they could
learn to
do that
that wouldn't
be a big

problem
but I guess
if I have
to use
two dimensions
together
to classify
objects
that would
have more
problems
because this
involves
considered
more
than one
dimension
simultaneously
I'm not sure
I'm making
sense here
this is
the Jonas
Langer
work
that he
did with
kids
and that's
a remarkable
work that
Jonas Langer
did
and then
there were
these people
in Rome
I think

it was
Poti
at the
lab of
Biopark
in Rome
that replicated
that with
chimpanzees
I think
it was
Spinozzi
who
replicated
it and
found that
chimpanzees
were more
compromised
in order to
have this
kind of
classificatory
relationship
relations
so
and I think
that's a
very remarkable
ability
that
we develop
so
I don't know
if I answered
your question
Michael
perhaps

that
comparison
of two
dimensions
color
and shape
and then
the
snakes
and ladders
overlay
that lets
us take
the hypotenuse
is that
abductive
jump
enabled
developmentally
and neuroepigenetically
by all
of these
different
features
which happen
over much
slower
timescales
especially
considering
the actual
millisecond
timescale
associated
with recognition
those other
changes are
so much

slower
but the
ability
to
move
diagonally
on the
grid
through the
rows
and the
columns
and turn
that into
movement
on a
manifold
is the
ability
looked at
from the
either
row
or column
perspective
to move
in both
at once
but by
condensing
those two
dimensions
into one
there can
be new
kinds of
cognitive
movements

and then
applying
that
cognitivist
lens
also
evolutionary
transitions
even
without
the
complex
cumulative
feedback
with the
eco-cultural
niche
merely
in the
abductive
moment
and just
one other
piece I
wanted to
return to
is that
every
variable
can have
a precision
estimator
or not
in the
generative
models
and active
inference

every variable
can be
fixed
and so
the model
can just
be
persistently
surprised
with no
updating
or
one can
have
any
kind
of
prior
or
hyper
prior
and so
on
and that
enables
learning
and that's
what enables
not just
estimation
of the
mean
or the
center
of gravity
the
expectation
of

but also
the variance
the whole
distribution
and then
it's
by applying
that kind
of
precision
on
everything
view
which as
Michael
brought up
is exactly
what modern
statistics
does
you can't
do
inference
with a
t-test
unless you
have the
mean
and the
variance
won't go
into any
more details
on the
statistics
but every
variable
has the

center of
gravity
mean
estimator
and the
variance
estimator
and so
we talked
about
precision
on
sensory
information
the
mapping
between
observations
and
hidden
state
estimates
and then
also
there's
this
action
oriented
precision
on how
things
will
unfold
and
you
might
be able
to

update
your
precision
on
sensory
evidence
by
seeing
something
different
but
updating
your
precision
on how
things
will
unfold
is
oriented
around
taking
actions
Daniel you
must have
some
questions
I can't
be the
only one
floating
questions
and having
my
zone of
bounded
surprise
being

expanded
here
I
have
questions
I can
ask a
question
or
Michael
or
Hector
or
Carl
please
feel
free
if you
hear me
I would
like to
pick up
with something
that
Professor
Christon
mentioned
before
and
that
was
kind
of
was
intriguing
because it
is true
that some

great
artists
like
Bangkok
and others
were very
creative
and also
I don't know
if it's
appropriate to
say they were
in the
verge of
psychosis
I don't know
if it's
appropriate
because I
don't have
any
statistics
but I'm
thinking
if being
innovative
is
this again
is misfiring
so I
would like
to
separate
two
things
one
is
how

innovative
you
are
and
I'm
thinking
in the
Econish
if you
are well
adapted
to an
environment
it's not
a good
idea
that you
try
different
things
all the
time
because
imagine
that you
have
some
place
where
there
are
nice
berries
you
have
discovered
a safe
path

to that
place
and
if
you
keep
to
that
path
you
go
to
the
berries
you
get
there
you
feed
so
that's
to
try
something
new
so
you
are
very
innovative
and
one
day
you
decide
to
try
another

path
to
the
berries
a

that the environment changes and if the environment changes significantly we are not well adapted because we don't have alternative behaviors that are adaptive to the new environment so i guess with this happens like psychological traits nobody has extremes so we don't have people with extreme psychological personalities i mean we have pathologies but the selection has favored kind of intermediate levels so i guess and professor friston can correct me on this he knows more about psychopathology but i guess it happens with innovation that we need some kind of balance and what this is doing for me is how humans escape that balance in terms of how they manage to be creative and innovative at times because we are very uh we repeat many routines so but still we can be very creative and we have selected so we have been successful enough to our econish and i'm just thinking in this content factors idea so i think this might have played a major role the ability to have counterfactors because perhaps instead of risking my life every time i innovate i can rehearse what will happen and as professor friston mentioned before if i have a good precision i can imagine things that will unfold if the genetic model is very deep i can think ahead many steps and see how i will fare in the future so i can kind of rehearse things in my head before i put them into practice in the real world and this might help me to be innovative but with a kind of bounded risk or control risk i don't know if this makes sense but i think it must be some kind of balance between if i can rehearse thing in my head i'm very good and predicting things that will um evolve or unfold many steps ahead i can see how i will fare and then i can reach to be innovative i don't know i mean i'm very i'm being very speculative here okay so i just want to make clear that i'm being very speculative please carl yeah no i'd like to speak to that yeah i don't yes you may be uh speculating uh but i think those um those observations are very veridical and very and very useful um i mean you've brought a number of fundamental issues to the table first of all the importance of being able to plan um to have a generative model and the consequences of actions is not the gift of all species and and in some instances um in your words when the environment doesn't change you that would be an undue complexity cost to carry around a brain with a generative model of the consequences of my actions if i don't need it if i was a virus or an insect i i wouldn't i wouldn't need that it would be um it would be very inefficient um so that sort of lends um brings you to a picture that our world is sufficiently complex and changing at pace the um committing to um very simple underlying expressive generative models is not an option and one really important way of making them more expressive is to increase that temporal horizon or temporal depth so that we can you know in accord with things like simulation theory roll out into the future and and now um become the kind of phenotypes that can select among a number of plausible policies um and you know i think that would be one bright line between lots of you know different genres and

different species you know that could be just um drawn on a dimension which is just the depth of planning that my generative model will support um and we've talked previously about uh that being constrained by the precision afforded in a given context um my beliefs about how i would roll out into the for into the future so i think that's just you know from the from my perspective at least that's a really important observation that um the depth of planning is a very important functional characteristic of the generative models that are implicit in phenotypic behavior and possibly in terms of um what um what kinds of models are subject to selective uh pressure at an evolutionary or indeed at your nevo-devo um at a nevo-devo uh level um the the other the other i think key insight that you um you bring to the table here um relates to um what affords the right balance between novelty seeking exploratory behavior um and the risks that that would entail and i think you know um that from a purely um sort of mathematical point of view is a really important observation um just to sort of join the dots with other ways of understanding how the universe works um that um particular question um is very easily framed in terms of constraints so the free energy principle that from which active inference inherits is um one way of or is dual to um something called the maximum entropy principle under constraints so that this was um

brought to physics in the 20th century by um a foundational thinker e.t james who basically rewrote physics in terms of probability theory and said that basically everything that um of interest at least or physically um conforms to a constrained maximum entropy principle so that actually is just an expression um or for um um or tends the free energy principle in the sense that the constraints um in from the point of view of free energy principle um are just what is brought to the table by a generative model um and um the entropy um pertains to the entropy of belief so you've got this um this dialectic um simply at the level of the physics of uh measurement and inference which means that i'm always trying to explain any measurement and um such that my predictions are as accurate as possible but at the same time um

leaving having the maximum latitude so that i don't over commit in accord with occam's principle so mathematically what that means is the entropy of my beliefs has to be as big as possible provided i provide provided i have an accurate um the con given the constraint that my account or my measurement has to be an accurate measurement so that that may sound if you like completely unconnected to whether i choose to um wander away from the berry path and but in fact exactly the same math

applies to these quantities either the constraint entropy or the variational free energy in the future and what happens is um you get exactly the same dialectic you have two pressures on the one hand to comply with occam's um principle or occam's razor and on the other hand to um um um to you to uh keep all your options open um which in a sense is exactly do i uh comply with the constraints um um that i know i'm going to get my berries and i'm not going to be eaten my myself by a predator or do i keep my options open and try that other path so this the this so dialectic is just bait into the physics of any self-organizing system um and practically um you know it emerges in a number of ways one way of writing down the free energy or the constrained entropy from james's point of view

expected given a particular policy do i go down this path or that path um is in terms of ambiguity and risk to you are using exactly your words you know uh so the risk is just the relative entropy or the divergence or discrepancy between what i think will happen and what i expect to happen to me so i expect not to be eaten and that's the key constraint at hand here so it's not about rewards anymore we're throwing away the notion of rewards and we're now replacing it with a multi-dimensional construct which is all the constraints in the ways of being like having a certain temperature or not being very cold or very hot not being eaten not being dead just keeping within my characteristic states so you write down those characteristic states in terms of constraints and those constraints basically preclude certain states of being and then i just compare well what will my states be if i do this and how far away from um how far away uh is that anticipation from my prior constraints and that that literally is called risk it's it's the kale divergence between your policy dependent um predictive posterior and your prior preferences that have the constraints the other thing is the ambiguity um and that simply reduces to um making sure that you make moves in a way that um in underwrites a um an unambiguous precise mapping uh between um what you see and um what caused your sensation so you avoid dark

rooms basically you avoid the dark you do avoid dark alleys you don't go uh where you can't see what's happening so that's the ambiguity so together the expected free energy is equal to risk plus ambiguity um and um when read like that then the balance between the risk and the ambiguity or another way of carving this up is in terms of expected information gain an expected cost or um um um its complement expected value where the value is just the uh how to what degree am i complying with my constraints um is um um enables you to to now write down the balance that resolves to a certain extent dissolves that dialectic by the precision of your prior preferences your constraints so if you're the kind of creature or child or insect um that has very precise beliefs about the kinds of outcomes i expect to anticipate and i may have very precise beliefs about my body temperature for example but i may have very less precise beliefs about where i am in terms of latitude i could be in alabama i could be in newcastle london london i can still be me but i can't be minus uh you know minus two degrees centigrade and still be me so some constraints are very precisely specified and other constraints are less precisely specified so the risk um is you know um is a is determined by um the precision of your prior beliefs about your characteristic states the precision of the constraints you bring to the table as a phenotype and something that will be innate and completely sub-personal just wide into your homeostasis and some will be learnable some may even be um sort of propositional you know we can talk about them um but whatever they have to be that because they define basically what kind of thing you think you are and how you act to act accordingly so the balance between the um you know whether to go down another path as opposed to the berry path is going to depend upon this particular creature or phenotype and specifically the precision of their prior preferences um in terms of what outcomes they would a priori um entertain and if they have very very precise um preferences over not being eaten and there's even a marginal small probability of being eaten if they stray from the current path then they won't explore if on the other hand they have a slightly more liberal um expectation um about um being uh about

the encounters with a predator for example um then there will be more um there will be more more likely to engage in novelty seeking and exploratory behavior so you know perhaps put this even more simply you know this is the exploration exploitation dilemma written in terms of the physics of self organization under planning um and you know and it all reduces again to precision and in this instance it's the precision of the constraints that um articulated as the uh your prior preferences your characteristic states your um the kinds of states that you you know you are um you believe characterize you that if you stray from you cease to be you cease to exist uh basically so i thought i was i thought that's i don't think what you said was speculation i think they're all fundamentals of behavior and actually um from an ethnological perspective you know they speak to the heart of the of choice behavior um and you know why we do the things we do um and why it is always in the background the game of saying do i explore or do i play it safe awesome thank you carl dean thank you muted daniel oh thank you thank you thank you carl go ahead dean i i i would i just like to run with what carl was saying this is mostly for hector and michael um

here's a here's a here's a case in point all the collective wisdom on the side of a volcano can suddenly be arrested and yet there is still a pompeii today i wonder what you gentlemen think that says about certain species zone of founded surprise

dean does this refer to the materia the fixed materiality alone of the ruins of pompeii of which no more will ever exist in the archaeological context being able to be unpacked into this collective wisdom i'm just curious because we are talking about regular irregularities and regular irregularities these these seismic events occur in the timeline and i'm just curious what we think then as we reflect on that and still adopt that kind of risk mindset as i said i just want to run with what carl was was suggesting and i i don't know that i don't know that there's an equivalent i don't know if i if i if if chimpanzees would do what humans have done like see the evidence and still you know park themselves right in the path of of danger i'm just curious what they think about that in terms of the the timeline aspect of this

is

hector

or michael go ahead

i think you're muted now hector

sorry my microsoft um so it's a it's a kind of difficult question and i'm not sure if i understood correctly um but as you're referring to the tone of bundle surprise and linking what what professor christian said before if i'm to take a new road which is i don't know what awaits for me there okay but if i'm very good at making guesses or inferring what might happen based on information i have gathered before so i'm good at anticipating what might unfold okay because i have this complex machinery let's call it working memory which is a construct i have a powerful working memory so i can entertain many possibilities and there is something that kind of ringed a bell when professor of wisdom was talking of precision which is the concept of metacognition so metacognition refers to thinking about my own thinking right so

i and you have to clear um to clear aspects one is

how sure am i about my knowledge for instance i'm sure i know

something even when i cannot remember this is the tip of the tongue effect so i can be or i can give you an answer and be 100 percent sure this answer is correct i can be 80 percent short 50 percent sure so i can evaluate how much sure i am about something okay so this i think it brings a bit like precision so if i make some expect plans and i think what might unfold i can be very sure this will happen or not and this probably has to do with my metacognition so how sure i am about what will happen and that might help me to decide whether i go or not um and coming back to your son of under surprise i'm not sure if i'm digressing but perhaps i can preempt the surprise if i have a good precision or good metacognition so if i'm sure that i can rehearse things in my head and see how they will unfold they won't be that surprising later so i i'm just speculating perhaps this ability to have content factors to have these kind of rehearsing in my head and this depth of generative models can be empty surprise and here i'm speculating clearly so i don't know if i answered your question about pompeii because i'm not sure that i understood but just bring some some ideas that might be useful but again they are speculative

it'd be nice just to link that notion of metacognition and having um beliefs about beliefs you know the precision of a belief about the likelihood of being eaten by a predator um i think that's what daniel was speaking to before when he was talking about um the hierarchical depth of models and their increasing complexification um one really important way of making making a model uh more expressive and context sensitive is to induce a hierarchy of exactly the sort that hector was talking about in terms of this metacognitive aspect but not just um content but crucially again coming back to the um the confidence and the uh and the precision um i think that's a really central concept but i was just thinking you know if you had to commit to what kinds of epigenetically specified constraints you would um associate with the genome what in what space would they would they would they live um and my guess is that most of it is about the constraints and within that it's going to be the constraints on the precisions that there are certain sort of innate ranges um of outcomes that i will entertain um and in in in in principle at least you can actually just write down an entire generative model in terms of of constraints and in terms of precisions in the sense that and if the generative model is universally expressive and everything goes then you can sort of carve out an apt uh structural form that um for that um generative model despite shrinking some of or 99.9 of this overly expressive generative model uh to zero with very very precise uh constraints so uh well perhaps i'll turn this into a question you know if if you uh uh trying to simulate um artificial life in its most generic sense if you're trying to simulate tool use for example the emergence of tool use in a synthetic population um and you had to actually write down

what could be passed via um genetically or epigenetically to each subsequent generation

what what kinds of things would you specify

that's kind of my question in the same in the same sense how do we where where do we draw those arbitrary lines i mean it's the it's the thing that comes up again and again as we continue to sort of

see the next the next sedimentary layer of papers discussing this right and but the but the fact that it can be moving and the fact that it it isn't con there is a rate constant that you mentioned in the last live stream that has to be um working functioning and effective and then there's still this question of so so what is our our what are we attending to and what are we still available to all at once and how do we organize that in a way that it most closely as you say precisely it that that's the best way to describe it what's really going on and that that's that that seems to me at least to be a moving target just one quick speculative mapping as well first the um instead of just stacking bricks with content on content on content intercalating layers of content estimation with metacognitive estimation is what allows the extendability of the multi-layer predictive processing type architectures it's actually like the back and forth or the back and forth or the interplay between the content estimators and the variance estimators that allows that model to not just be described in terms of its center of gravity but also in terms of its full distribution and then to carl's question about what is being passed down genetically and epigenetically and thinking about those terms even generically like in the context of perhaps source code and a configuration file or some other mapping of like what is inscribed from a classical information genetic perspective and then what is contextually and more rapidly transmitted there's an interesting angle i could see it this way or potentially the opposite but there's a shorter term epigenetic firing rate and then a slower updating of neuromodulatory like genetic possibilities kind of opening and closing valves over a slower time scale and then of course these are nested within another one so

what is what is neuromodulatory this can also itself be a firing rate in other ways but just focusing on this direction the uh inscribed constraints are like the width of the aperture and then the actual flow of the the photons through the lens is an instantaneous firing rate that's happening way faster than the aperture can be

dilated or not and so merely the fact that there are multiple concurrent time scales and the associated interplay of their rate constants tells us a lot about this kind of learning architecture

what we do with that understanding is the action question and related to our preferences and precisions however it seems pretty key that to alternate estimates about content with metacognitive type estimates and to package that set of two belief distributions mean and variance to package those

also as one so it can be handled as a composite one

and just to point out that um learning rates are the manifestation of that second order specification that uh precision so you know they are the same thing you know at every um at every time scale right so you could you could articulate the um the developmental stages for example the um the not use word rejuvenation but certainly you know the the um the changes say in late adolescence um as a programmed learning rate or learnability that can be read at some level as a precision because the precisions are just the rate constants that link the state to the rate of change of the state um and that basically is just a a learning rate at many many different levels so i think there's a a sort of uniform um way of putting together these different notions some useful notions here um um uh from machine learning are um sort of the notion of learning to infer you know so i was just thinking

about my answer to my own question and clearly you can't write down a generative model that's going to be able to um get up and go to school in the morning uh all you can do is to um create a self-organizing blob that has exactly the right learning rates so that it has exactly the right learnability so that it can acquire a generative model and learn through experience dependent plasticity that model of its little lived world of course that little lived world has been constructed by other grown-up babies so you know we get back to this circular causality that evolution is trying to provide you with the right ability learning rates precisions and structural constraints on your generative model but your generative models from evolution's point of view are actually creating the environment that has to be learned so there's to my mind there's this you know a very beautiful circular causality here that you know as fast as you're uh learning um you're actually creating new worlds in which to learn you know i guess we've seen that at the moment with the advances in technology and the like which brings me back to my question about the sort of the top-down versus bottom-up causation um you know what is a good model of this world what is a good phenotype in our final 15 minutes or so here how do we move beyond the 53.2

the other way to the author's view either to the author's their directions and excitements about the work or anyone's exhortations or raising of possibilities for other ways it can be developed in our final 15 minutes or so here how do we move beyond the fact that we can see that we can see

and i wonder if it's worth bearing in mind that increasing technological not necessarily complexity but at least variability this may well have permitted a reduction in selection pressure on the phenotype and evidently affecting uh both the genotype and epigenetic uh variability i say this because i would like to draw attention to the the graph which i presented last week which was taken from professor philip tobias's work on early humans showing an increasing range of variation in the cranial capacity of uh bipedal hominids from the osteopithecus signs through to homo sapiens suggesting a reduction in um selection pressure for particular um well for the size of the brain basically uh i find it hard not to relate this to the advantages that slowly accrued to uh tool use and thereby of course the variability and variation of tool making um thereby reducing the risk the risk or some of the risks uh existential risks that otherwise plague or have plagued other hominins whether extant where the ancestors of extant great apes or indeed the earlier forms of uh bipedal hominins or palaeoanthropus signs panthropus um kenianthropus uh etc uh well who knows but it's it's perhaps um uh not totally irrelevant that this uh variability uh has grown over the past couple of million years and is now very wide as i mentioned uh last week

this i think is in contradistinction to the the way in which selection pressures have been highly
um discriminatory um discriminatory in for example producing let's say the wide range of
darwin's famous galapagos finches yeah just to give a well-known example
thank you carl

um coming back to this theme of um selection for selectability um and the notion of um precision or
learning rates cast in terms of um mutation rates so you're my favorite example of and i'm sure there
are better ones nowadays when i was reading around this subject my favorite example was um the story
told by ernst to mayor about drosophila fruit flies being reared in volatile versus non-volatile
environments so you're rearing generation after generation at a fairly fixed temperature and then
an equivalent uh set of generations where they varied the temperature um you know a few degrees plus and
minus um and what they showed was that the mutation rate of the drosophila in the volatile
changing environment was greater than the mutation rate the genetic mutation rate in the less volatile
population or transgenerational populations which i thought was a beautiful example of selection for
selectability itself that you know that um increased variation um certainly or it's you know phenotypic
expression of implicit uh increased genomic variability was itself an adaptive response to to the rate of
change of the environment so my question is is there a similar sort of picture or view of um variation in
either phenotypic or genetic um characteristics in um over a longer time scale in in sort of um human
evolution

uh and if there is is there any evidence um that there's going that there is in the past century or so um
perhaps this is a bit of an ambitious uh ask but uh is there any evidence that the the the there has
been a sort of phase transition in the variability and the evolvability and the associated um phenotypic
variation

the empirical or an empirical research agenda would be to make a hierarchical active inference model
and perform dynamic causal modeling on different variability characters across levels of phenotypic and
genotypic

integration and have that model in use

but a very provocative and interesting question carl

and to what extent is it happening with our cellular architecture and to what extent

are these processes happening with our niche given these unique cumulative cultural generative model
generative process thinking through other minds increasingly the production of artifacts that enable
not just offloading but outsourcing

not just offloading these do represent unique cognitive niches and so

it will be quite interesting to investigate

perhaps if we each want to have some closing round thoughts

perhaps dean with the first word thank you because you know i hate having the last one

i'm just curious gentlemen thank you first again for for your generosity i'm just curious now that
you've dipped your toes in this this uh active inference institute live stream world what are
your what are your impressions is was it was it it was it helpful at all

and i think it was a very useful and mostly very fun i mean i've enjoyed a lot to have this discussion with all of you so it's been very enjoyable um and just for the future um i'm just wondering if there is such a thing like pre-empting surprise based on these counterfactuals i guess maybe it's already written and discussed and it's all clear but i wonder that this might be an interesting idea to to explore if i can preempt the surprise before i act i mean before even acting but that's kind of maybe that's resolved already anyway it has been fun thank you all honestly thank you that makes me think about saying i don't mean to surprise you but that's actually going to increase it's like oh boy has someone been injured but what is the show not tell of preempting surprise so that with a linear string in time there is the path from the old berry patch to the new berry patch such that it appears to be within a nested hierarchical model the path of least action to follow though in a less nested model it would be quite surprising

michael or carl you're kind of closing thoughts or questions as we bring this set of 53 to a close well i'll let michael have the last word um and uh just speak to uh you what you just said in uh hector's um comment i i think there's a lot of interest in that in in that notion you know sort of trying to uh preempt uh surprise i mean a simple reading of risk minimization you're just basically not engaging in policies that are going to result in surprising outcomes where surprise is is just the violation of uh prior preferences um so at that level i think that's that that's that sort of um um almost a truism um but there is an interesting um there's an interesting sort of implication of that which leads to notions of um the way that we choose to behave epistemically in terms of avoiding information that would be very surprising um sometimes articulate in terms of optimism bias sometimes in terms of wishful thinking but the kind of paradox that you get when we are base optimal creatures and yet we don't want to open the letter from the gp giving us our most recent cancer results or we don't want to open that letter whether we failed to get that job or pass that exam or indeed the anatomy dissection viva and there are certain things that we avoid there are certain surprises that we avoid or split another way there are certain ways of pre-empting surprise by avoiding knowing stuff that we might find surprising um and that's that's an emerging literature now which which takes you a little bit outside your your your comfort zone but it's a fascinating literature because you you'll find things like trends in cognitive science and and the like uh in answer to to dean's question about um these sessions because this is not my first session so but um i this but this does exemplify it yeah it's for fun and we're amongst friends and indulging ourselves and those people who care to listen to us so i think the next thing to do is is the next horizon although i do think dynamic calls and modeling of um evolutionary um trajectories is an excellent idea i i suspect the heavy lifting now is to get the ideas that we've been talking about out there i think that's what michael and hector have to do basically it's just keep keep on telling the story and in my experience usually takes about five to ten years before people start listening

thank you carl michael with the last word

well there have been these sessions have been fun and most uh illuminating there is a lot to be done on the matter of whether there have been phase transitions in the evolution of what one might call

evolvability in the genus homo over the past two million years there are big arguments amongst human evolution specialists between whether there was a unitary evolutionary um uh whether a unitary evolutionary model is acceptable or as uh lumpers like myself would like to believe or whether in fact there was a big um bush with a lot of dead end separate uh unsuccessful um species like homo florisiensis um homo uh lusoniensis

or even perhaps neanderthals uh with whom i've had a lot of contact uh having excavated about 15 of them um and these arguments are highly relevant to the matter of uh uh cognitive evolution and indeed uh as to whether uh the record of artifacts and tools uh can indeed uh inform us and i believe it can about uh just how um and how slowly i would say uh evolution proceeds proceeded

um of course the splitters believe it happened in in leaps jumps and leaps um i'm rather convinced darwinist i believe that things happen very very slowly and that we have to concentrate and with this i can conclude uh on the especially the neuroepigenetic um uh modification of uh genetic expression uh as uh affecting not merely morphology and physiology but also i would say um cognition and uh in the matter in hand of course um uh the ability or or not to uh that our surprisals um affect uh onlookers and other people uh permitting um let's say modification of behavior uh in a very wide in a far wider uh with far wider results than those that occur in uh many other uh species um including some of those of our closest uh our closest permanent um relatives and thank you very much for organizing this session daniel and dean it's been it's been a pleasure to take part i'm sorry that i have this terrible uh appearance but i

i really can't do much about that at the moment

i hope well last will be a few days

thank you all michael at least you have an excuse i have no excuse at all

all right till next time long live the mean long live the variance thank you bye everybody

thank you

Thank you.